

AI-Driven Distributed Fog Computing Architecture

Technical Project Sheet

Network Engineering – Fog Computing – Artificial Intelligence

1. Project Overview

Project Title: AI-Driven Distributed Fog Load Balancing & Anomaly-Aware Routing

Objective: Design and implement a **distributed fog computing architecture** capable of:

- Dynamic load balancing
- Network anomaly detection
- Intelligent routing decisions
- Performance comparison with classical fog architectures

2. Problem Statement

Traditional fog architectures suffer from:

- Centralized fog controller (Single Point of Failure)
- Static routing mechanisms
- Poor scalability under dynamic load
- Lack of intelligence for proactive decision-making

This project addresses these limitations by integrating **Artificial Intelligence directly into fog nodes**.

3. Chosen Architecture

- **IoT Layer:** Traffic generators and simulated sensors
- **Fog Layer:** Distributed fog nodes interconnected in mesh topology
- **Cloud Layer:** Monitoring, historical analysis (non-critical path)

Architecture Type: Distributed Fog (Decentralized Control)

4. Technology Choices

4.1 Simulation and Networking

Component	Technology
Fog Simulation	Fogbed (Containernet + Mininet)
Virtualization	Docker
Switching	Open vSwitch
SDN Controller	Ryu (Python)
Network Control	OpenFlow
Fog-to-Fog Communication	gRPC over TCP
IoT Messaging	MQTT

4.2 Artificial Intelligence Stack

Function	Model / Tool
Load Prediction	LSTM (PyTorch)
Anomaly Detection	Autoencoder (Scikit-learn)
Routing Optimization	Reinforcement Learning (Q-Learning)
Data Processing	NumPy, Pandas

5. Functional Requirements

- Dynamic traffic redirection between fog nodes
- Automatic detection of overload and anomalies
- Real-time routing reconfiguration
- Fault tolerance and fog node isolation

6. Non-Functional Requirements

Criterion	Requirement
Latency	Lower than centralized fog
Scalability	Dynamic addition of fog nodes
Resilience	No Single Point of Failure
Observability	Metrics, logs, dashboards
Reproducibility	Automated test scripts

7. Comparative Study

Metric	Centralized Fog	Distributed Fog (No AI)	Distributed Fog + AI	
Latency	High	Medium	Low	
Packet Loss	Medium	Low	Very Low	
Fault Tolerance	Low	Medium	High	
Reaction Time	Slow	Moderate	Fast	
Bandwidth Usage	Inefficient	Moderate	Optimized	

8. Test Scenarios

1. Progressive traffic load increase
2. Sudden overload on a fog node
3. Fog node failure
4. Network congestion simulation
5. Anomalous traffic injection

9. Monitoring and Visualization

- Prometheus for metrics collection
- Grafana dashboards for real-time visualization
- JSON logs for event analysis
- Mininet CLI for topology inspection

10. Expected Outcomes

- Reduced latency and packet loss
- Improved system resilience
- Intelligent fog behavior
- Clear justification of AI integration

Keywords: Fog Computing, Distributed Systems, SDN, Artificial Intelligence, Network Optimization